ANDREW HAGGSTROM

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA — May 2028
M.S. in Robotic Systems Development (MRSD)

Washington University in St. Louis, St. Louis, MO — May 2026
B.S. in Biomedical Engineering & Computer Science, GPA: 3.97
Relevant Coursework: Robot Mobility, Deep Learning, Robot Autonomy, Computer Vision, Probability & Statistics

TECHNICAL SKILLS

- **Robot Simulation:** MuJoCo, F1TENTH Gym (ROS 2), LIBERO manipulation benchmark
- **Languages:** C++, Python, CUDA, Bash/Shell, SQL, MATLAB
- **Robotics & Perception:** ROS/ROS 2, LiDAR-camera sensor fusion, frustum projection, point cloud processing, object detection and tracking, camera calibration, PID and reactive control
- **Platforms & Tooling:** Jetson Orin, Linux (Ubuntu), Docker, SLURM, Git, PyTorch, TensorRT, OpenCV

ROBOTICS PROJECTS AND RESEARCH

Autonomous Racing Perception Stack *C++, Python, ROS, TensorRT, Jetson Orin, LiDAR*
- Led perception for a Formula Student autonomous vehicle, implementing the detection-to-localization path in C++ ROS nodes that fed the planning and controls stack on a physical car.
- Fused camera-based cone detection with LiDAR point clouds through frustum projection and DBSCAN clustering, recovering metric 3D obstacle positions from 2D image detections and raw sensor returns.
- Defined ROS node interfaces and containerized the stack with Docker so mechanical, electrical, and controls subteams could integrate against a stable, reproducible specification.

Geometry Tokens in OpenVLA-OFT: Simulation-Based Policy Evaluation *MuJoCo, LIBERO, PyTorch, CUDA*
- Forked OpenVLA-OFT to inject a PointNet-encoded geometry token into a 7B vision-language-action policy, testing whether explicit 3D structure improves manipulation robustness.
- Created and ran a closed-loop evaluation campaign in simulation across 7 perturbation categories on the LIBERO manipulation benchmark, tracking training and evaluation runs in Wandb.
- Identified a reproducible dissociation between the standard offline metric and simulated task success, where success collapsed from 80% to 0% while offline error continued improving.

Automated Stave Pallet Counting *YOLO, FastAPI, ONNX, Docker*
Computer Vision Engineer, Engineers Test Kitchen (WashU engineering consulting)
- Built and deployed a vision system for a cooperage client to replace manual counting of 600 to 700 wood staves per pallet, a labor-intensive step performed by hand on the factory floor.
- Designed a two-stage YOLO pipeline in which a localizer model isolates the target pallet from cluttered background staves, then a counter model detects and counts individual staves within the cropped region.
- Shipped a containerized web application with a FastAPI backend and ONNX runtime inference, deployable on client hardware.

EXPERIENCE

ZS Associates — Seattle, WA
Data and Decision Associate Intern — May – August 2026
- Shipped a production natural language-to-SQL service in Python over 50+ relational tables, letting non-technical users self-serve analytics without engineering support.
- Developed distributed data pipelines in SQL, Spark, and Azure Data Lake across 5 cross-functional teams, and automated validation over 8M+ record mappings to catch defects before deployment.

MediBeacon — St. Louis, MO
Research and Development Intern — May 2024 – August 2024
- Automated safety and verification test analysis for prototype hardware with Python and MATLAB, working with hardware engineers to turn raw benchtop sensor data into pass/fail validation evidence.
- Replaced a manual measurement review process with scripted analysis, accelerating design-control reporting under FDA QSR 820.30 and flagging anomalous prototype behavior ahead of design reviews.

Medical Artificial Intelligence Lab, WashU — St. Louis, MO
Undergraduate Researcher — January 2025 – May 2026
- Built deep learning and gradient-boosting pipelines in PyTorch and scikit-learn on high-dimensional data, designing cross-validation protocols to produce generalization estimates.
- Designed feature-selection experiments to separate reproducible signal from selection artifacts under severe sample constraints, validating candidate findings against prior literature with faculty researchers.

Stryker — Mahwah, NJ
Business Solutions Engineering Intern — May 2025 – August 2025
- Automated a software approval workflow in Azure DevOps and engineered a data pipeline that cut turnaround time by 20% and improved throughput by 40%.
- Profiled the existing manual process to locate bottlenecks before rebuilding it, and ran daily Agile requirements sessions to scope a shippable backlog across engineering and business stakeholders.

LEADERSHIP

Washington University in St. Louis, *Teaching Assistant, Intro to Machine Learning; Probability & Statistics* August 2023 – May 2026
- Taught ML and probability through Python assignments, held weekly office hours, and debugged student code across two engineering courses.

Engineers Without Borders, *Lead Project Manager* — November 2023 – May 2026
- Directed 3 data analyst teams and coordinated with the US EPA, UN Environment Programme, and Missouri River Cities Initiative to model Mississippi River plastic pollution.